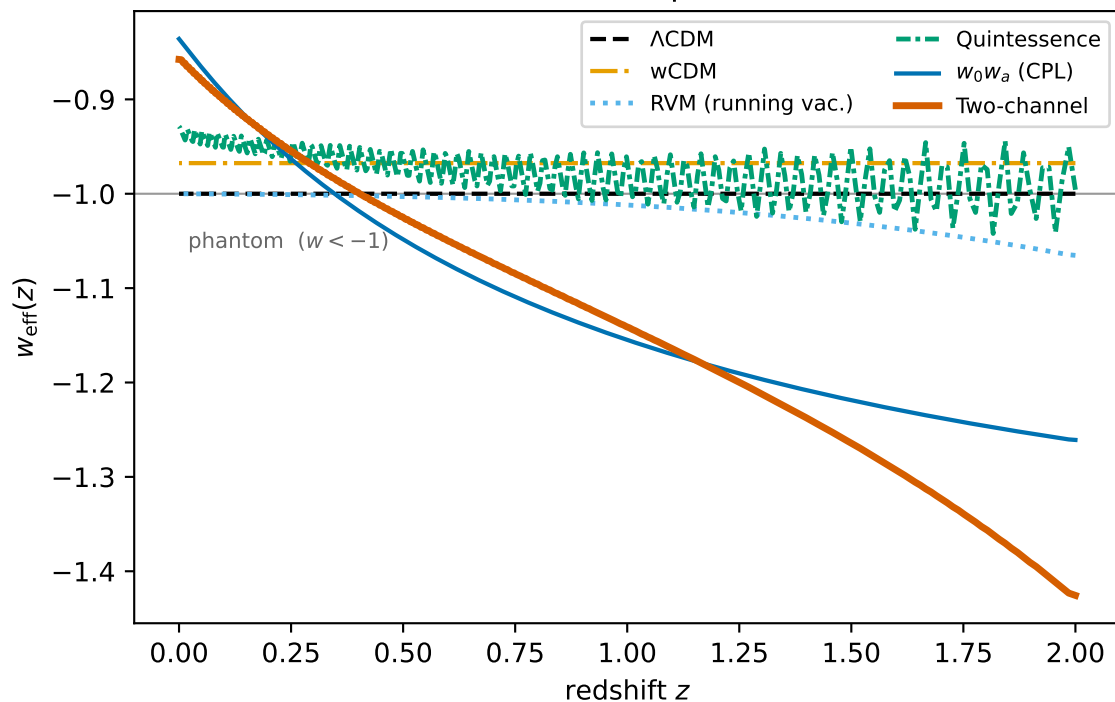
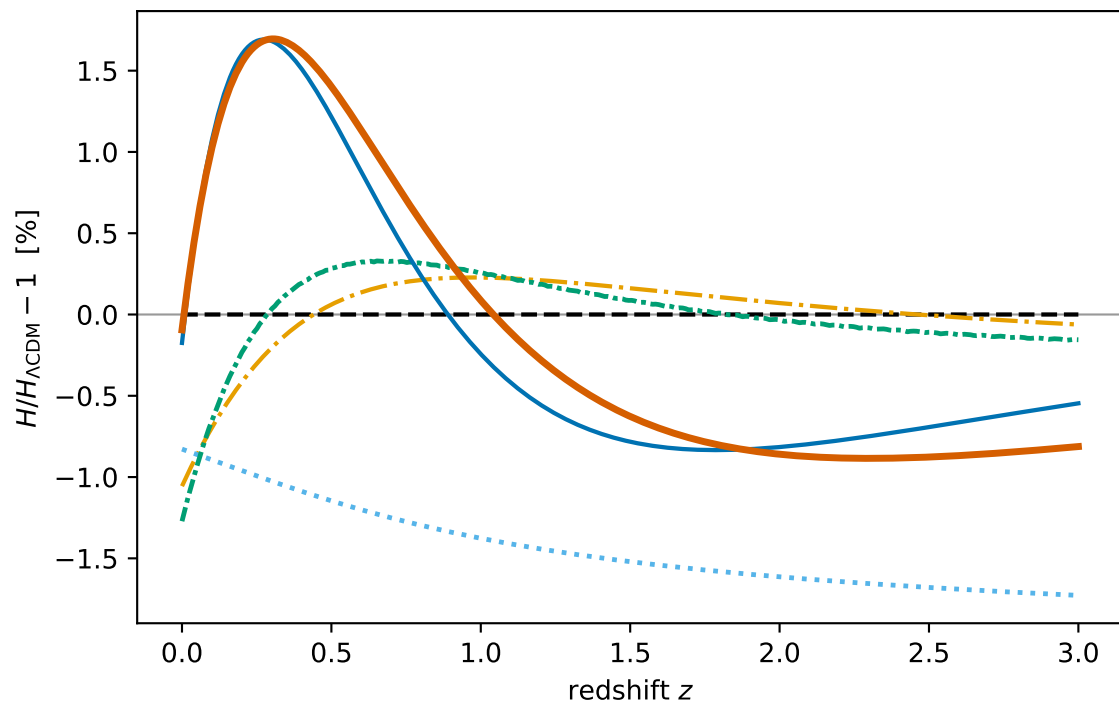


Benchmark suite: two-channel mechanism vs standard dark-energy models
(all fit on Planck distance priors + DESI DR1 BAO + Pantheon+)

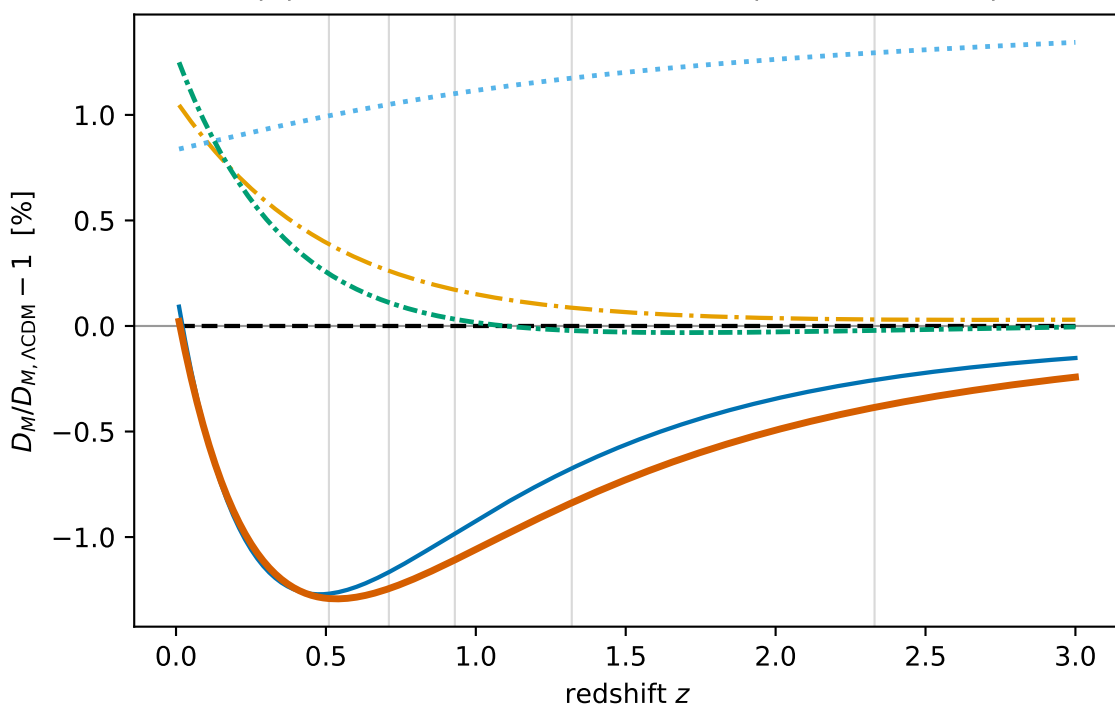
(1) Reconstructed equation of state



(2) Expansion-rate residual vs Λ CDM



(3) Distance residual vs Λ CDM (DESI z shaded)



(4) Growth $f\sigma_8(z)$ (GR, no DE clustering)

